

All the functions we have dealt with thus far have expressed one variable in terms of a function of the other, for example

$$y = \sqrt{x^3 + 1}, \quad y = x \sin x$$

In general, $y = f(x)$. However, we often want to consider more interesting curves where the relation is defined implicitly rather than explicitly. Such as

$$x^2 + y^2 = 25$$

$$\text{or, } x^3 + y^3 = 6xy$$

Sometimes we can separate these equations into our usual explicit forms (often multiple equations)

$$x^2 + y^2 = 25 \begin{cases} \rightarrow y = \sqrt{25 - x^2} \\ \rightarrow y = -\sqrt{25 - x^2} \end{cases}$$

Individually we can find the latter two derivatives and find

$$y' = \frac{-x}{\sqrt{25 - x^2}}, \quad y' = \frac{x}{\sqrt{25 - x^2}}$$

However, its not always easy to separate the implicit function.

$$x^3 + y^3 = 6xy$$

doesn't break apart nicely. So how do we find slope?

Implicit Differentiation:

If we treat y as a function of x , $y = f(x)$ then we can derive both sides of an equation to find a single expression for $\frac{dy}{dx}$.

* Since y is a function of x , $y' = \frac{dy}{dx}$ is a function but x is just a function of itself, so $x' = \frac{dx}{dx} = 1$.

Ex: Find equation of tangent to circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

1) Take $x^2 + y^2 = 25$ and differentiate both sides with respect to x , remember y is a function of x .

$$\begin{array}{lcl} \frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = \frac{d}{dx}(25) & : & \frac{d}{dx} y^2 = 2y \frac{dy}{dx} \\ \downarrow & : & \text{from Chain Rule} \\ 2x + 2y \frac{dy}{dx} = 0 & : & \end{array}$$

2) Solve for $\frac{dy}{dx}$:

$$2y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

3) Find slope at $(3, 4)$

$$\left. \frac{dy}{dx} \right|_{(x,y)=(3,4)} = -\frac{3}{4}$$

4) Find Tangent Line

$$y - 4 = -\frac{3}{4}(x - 3) \rightarrow y = -\frac{3}{4}x + \frac{25}{4}$$

$\frac{9}{4} + \frac{16}{4}$

Ex: Consider $y^3 + x^3 = 6xy$

a) Let's find y' (aka $\frac{dy}{dx}$)

$$\frac{d}{dx}(y^3) + \frac{d}{dx}(x^3) = \frac{d}{dx}(6xy)$$

↓
product Rule

$$3y^2 \frac{dy}{dx} + 3x^2 = 6\left(y + x \frac{dy}{dx}\right)$$
$$= 6y + 6x \frac{dy}{dx}$$

Solve for $\frac{dy}{dx}$:

$$3y^2 \frac{dy}{dx} - 6x \frac{dy}{dx} = 6y - 3x^2$$

$$\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2$$

$$y' = \frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$$

b) Eqn of tangent at $(3,3)$

$$\left. \frac{dy}{dx} \right|_{(3,3)} = \frac{2(3) - 3^2}{3^2 - 2(3)} = \frac{-3}{3} = -1$$

$$y - 3 = -(x - 3) \rightarrow \underline{y = -x + 6}$$

c) when is tangent horizontal?

$$\frac{dy}{dx} = 0 = \frac{2y - x^2}{y^2 - 2x} \leadsto 2y - x^2 = 0$$

$$y = \frac{x^2}{2}$$

↳ plug into original eqn

$$y = \frac{x^2}{2} \searrow$$

$$\left(\frac{x^2}{2}\right)^3 + x^3 = 6x\left(\frac{x^2}{2}\right)$$

$$\frac{x^6}{8} + x^3 = 3x^3$$

$$x^6 - 16x^3 = 0$$

$$x^3(x^3 - 16) \rightarrow x = 16^{1/3} = 2^{4/3} \text{ or } x = 0$$

$$x = 0, y = \frac{0^2}{2} = 0$$

$$x = 2^{4/3}, y = \frac{(2^{4/3})^2}{2} = \frac{2^{8/3}}{2} = 2^{5/3}$$

So, at $\left(2^{4/3}, 2^{5/3}\right)$ tangent is horizontal

at $(0,0)$ $\frac{dy}{dx} = \frac{0}{0}$, so we don't know (in fact there is a vertical and horizontal tangent)

Ex: $\sin(x+y) = y^2 \cos(x)$, find y'

$$\frac{d}{dx} \sin(x+y) = \frac{d}{dx} y^2 \cos(x)$$

$$\cos(x+y) \left(1 + \frac{dy}{dx}\right) = 2y \frac{dy}{dx} \cos(x) - y^2 \sin(x)$$

$$\frac{dy}{dx} \cos(x+y) - 2y \frac{dy}{dx} \cos(x) = -(y^2 \sin(x) + \cos(x+y))$$

$$\frac{dy}{dx} = - \frac{y^2 \sin(x) + \cos(x+y)}{\cos(x+y) - 2y \cos(x)}$$

* Check on graph at $(0,0)$ slope is -1 .

Ex: Find y' for $\cos(x - \sin y) = \sin(y - \sin x)$

$$\frac{d}{dx}(\cos(x - \sin y)) = \frac{d}{dx} \sin(y - \sin x)$$

$$(-1 + \cos(y) \frac{dy}{dx}) \sin(x - \sin(y)) = \left(\frac{dy}{dx} - \cos(x) \right) \cos(y - \sin(x))$$

$$\frac{dy}{dx} (\cos(y) \sin(x - \sin(y)) - \cos(y - \sin(x))) = \cos(x - \sin(y)) - \cos(x) \cos(y - \sin(x))$$

$$\frac{dy}{dx} = \frac{\sin(x - \sin(y)) - \cos(x) \cos(y - \sin(x))}{\cos(y) \sin(x - \sin(y)) - \cos(y - \sin(x))}$$

Ex: $\sin(xy) = \sin(x) + \sin(y)$

$$\frac{d}{dx} \sin(xy) = \frac{d}{dx} \sin(x) + \frac{d}{dx} \sin(y)$$

$$\left(y + \frac{dy}{dx} x \right) \cos(xy) = \cos(x) + \frac{dy}{dx} \cos(y)$$

$$\frac{dy}{dx} (x - \cos(y)) = \cos(x) - y \cos(xy)$$

$$\frac{dy}{dx} = \frac{\cos(x) - y \cos(xy)}{x - \cos(y)}$$

Second Derivatives:

Finding y'' is still possible, but much more difficult.

Find y'' if $x^4 + y^4 = 16$

Recall $y' = \frac{dy}{dx}$

$$\frac{d}{dx} x^4 + \frac{d}{dx} y^4 = \frac{d}{dx} 16$$

$$4x^3 + 4y^3 y' = 0$$

$$y' = -\frac{x^3}{y^3}$$

For y'' we derive again with quotient Rule.

$$\frac{d}{dx} y' = \frac{d}{dx} \left(-\frac{x^3}{y^3} \right)$$

$$y'' = -\frac{3x^2 y^3 - 3y^2 x^3 y'}{y^6}$$

We don't want y' here since an explicit formula is much nicer.

So, Substitute $y' = -\frac{x^3}{y^3}$

$$y'' = -\frac{3x^2y^3 - 3y^2x^3\left(-\frac{x^3}{y^3}\right)}{y^6}$$

$$= -\frac{3x^2y^3 + 3\frac{x^6}{y}}{y^6}$$

$$= -\frac{3x^2(y^4 + x^4)}{y^7}$$

Recall $y^4 + x^4 = 16$

$$y'' = -48 \frac{x^2}{y^7}$$